RAMDANI KHALED

AI Systems Builder | Agent Architecture | Data Engineering

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Open to Master's opportunities in Europe. | https://portfolio-seven-alpha-fjhei5q53o.vercel.app/

PROFESSIONAL SUMMARY

Final-year Computer Science student at Universite Badji Mokhtar, Annaba. Self-taught specialist in AI agent architecture and data engineering. Designs and deploys fully local, privacy-first AI systems using Ollama with zero cloud dependency and zero API cost. Active member of the government-recognized AI and Robotics Association of Annaba Wilaya. Currently interning at ANEM (Agence Nationale de l'Emploi) building a national employment intelligence platform. Mission: build AI systems that deliver real-world impact, then build a startup around them.

EDUCATION

Bachelor of Science in Computer Science
Universite Badji Mokhtar, Annaba, Algeria

Expected: 2026

- Final-year student with 3 years of English-medium instruction
- Member, AI and Robotics Association of Annaba Wilaya (government-recognized)

EXPERIENCE

AI Systems Intern — ANEM (Agence Nationale de l'Emploi)
National Employment Agency of Algeria | Annaba, Algeria

2026 (January) – Present

- Designing and building a national employment intelligence platform using multi-agent ML architecture
- Architected a NoSQL Big Data pipeline across 4 data streams: Profiles, Offers, Placements, and References
- Developed a KNN-based recommendation engine for skill gap analysis and job matching
- Implemented dynamic weight optimization using supervised learning trained on historical placement data

PROJECTS

ARIA — Adaptive Reasoning and Intelligence Architecture

In Development

Tech: Python, LangGraph, Ollama, ChromaDB, Streamlit, NetworkX

- Multi-brain AI analytics agent with a three-brain cognitive loop: Planner, Advisor, and Meta-Evaluator
- Hybrid local/cloud execution with per-step data sensitivity routing and cryptographic SHA-256 audit trails
- Three-layer memory system: Working Memory, Episodic Memory (ChromaDB), Strategic Memory (NetworkX knowledge graph)

ANEM AI Engine — Intelligent Employability Scoring System

Active — Government Internship

Tech: MongoDB, Elasticsearch, scikit-learn, Python, NoSQL

- Multi-agent ML system for national employment analytics processing large-scale workforce data
- NoSQL Big Data pipeline spanning 4 streams with KNN-based skill gap recommendation engine
- Dynamic weight optimization via supervised learning on historical placement records

Local AI Desktop Agent

Explored & Configured

Tech: Ollama, UI-TARS (ByteDance), Python, macOS, qwen2.5-coder:7b

- Deployed and configured UI-TARS (ByteDance) locally on macOS, connected to Qwen2.5-coder:7b via Ollama runtime
- Achieved fully offline desktop automation with zero cloud dependency and zero API cost
- Explored local LLM integration patterns for autonomous desktop task execution on consumer hardware

Wyag Lite — Git Reimplementation in Python

Complete

Tech: Python, Systems Programming, Git Internals

- Rebuilt Git internals from scratch in pure Python with zero external dependencies
- Implements repository objects, commits, trees, refs, tags, and log commands faithful to Git specification

TECHNICAL SKILLS

- **AI & Agents:** Multi-agent architecture, LLM orchestration, local inference (Ollama), LangGraph, RAG, vector stores, agentic pipelines
- **Machine Learning:** scikit-learn, NumPy, Pandas, OpenCV, MediaPipe, supervised learning, KNN
- **Data Engineering:** MongoDB, Elasticsearch, NoSQL design, ChromaDB, Pinecone, stream processing
- **Development:** Python (primary), JavaScript, C, Java (basic), Git, Streamlit, REST APIs

LANGUAGES

- **Arabic:** Native
- **English:** B1-B2 — 3 years of English-medium university instruction

- **French:** Intermediate

MEMBERSHIPS & ACTIVITIES

- Member, AI and Robotics Association of Annaba Wilaya — government-recognized organization promoting AI innovation in Algeria
- Builder focus: privacy-first, fully local AI systems — zero cloud, zero API cost architecture philosophy